

Abid Karim

Backend & AI Engineer | Node.js, Python, RAG Systems | LUMS CS+Finance '26

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EXPERIENCE

10Pearls

Software Engineer Intern

Lahore, Pakistan

Sep 2025 – Nov 2025

- Built a real-time full-stack TypeScript MERN notes application with advanced filtering and rich-text editing features
- Implemented secure JWT authentication via HTTP-only cookies and WebSocket-based real-time notifications
- Designed OpenAPI-compliant RESTful APIs, containerized services using Docker, and integrated SonarQube in CI/CD pipelines, resolving **12 high-severity security findings** before production deployment

AI Maternal Health Platform

Team Lead – Final Year Project (LUMS)

Lahore, Pakistan

Sep 2025 – April 2026

- Led a **5-member** team to build a production-grade AI maternal health platform using FastAPI, React, and PostgreSQL
- Designed a **multi-tenant RAG system** using Google Gemini API and ChromaDB with semantic chunking strategy; built a database AI agent to handle natural language maternal health queries
- Engineered dual data architecture (PostgreSQL + ChromaDB) and implemented **CI/CD pipelines** with Docker and Kubernetes for scalable, containerized deployment

LUMS

Teaching Assistant – Software Engineering

Lahore, Pakistan

Jan 2026 – April 2026

- Mentored **300+** students in Java and Android development; conducted code reviews for **50+** projects and managed Git/GitHub workflows
- Ran targeted workshops and office hours resulting in measurably higher project submission and completion rates
- Developed automated grading scripts in Python that cut assessment turnaround time by **40%** while maintaining evaluation consistency

Wetual – NIC Lahore

Business Analyst Intern

Lahore, Pakistan (Hybrid)

Jun 2023 – Aug 2023

- Built Python and SQL data pipelines to analyse financial datasets; surfaced pricing insights that informed early-stage product decisions, leveraging CS+Finance background
- Supported drafting of **SAFE agreements**, contributing to legal and financial documentation for fundraising processes
- Conducted primary market research across **50+** startups to validate product-market fit and competitive positioning

PROJECTS

CampusCaravan – Carpooling Platform | Node.js, TypeScript, React, PostgreSQL, WebSockets

Jan 2024 – May 2024

- Led team of **5** to build and ship a carpooling platform adopted by **150+** students at LUMS; backend built with Node.js, TypeScript, and PostgreSQL
- Implemented real-time chat via WebSockets, AI content moderation with Gemini API, and JWT authentication with HTTP-only cookies
- Optimized N+1 database queries and indexing strategy, measurably improving search and feed response latency

LUMSPiRE – Visual Discovery Platform | MERN Stack, TypeScript, WebSockets

Jun 2025 – Jul 2025

- Built a Pinterest-style visual discovery platform with personalized feeds, serving **100+** users with average page load under **1.5s**
- Integrated WebSockets for real-time messaging and notifications; implemented JWT authentication with HTTP-only cookies
- Optimized image loading with lazy loading and CDN caching strategy, significantly reducing asset delivery time

SKILLS

Languages: TypeScript, Python, SQL, C++, Java

Backend & Infrastructure: Node.js, Express.js, FastAPI, PostgreSQL, Docker, Kubernetes, CI/CD, Git

Frontend: React.js, HTML/CSS

AI Stack: RAG Systems, AI Agents, LangChain, ChromaDB, Google Gemini API, Pandas

EDUCATION

Lahore University of Management Sciences (LUMS)

BSc Computer Science and Finance

Lahore, Pakistan

Sep 2022 – May 2026

- **Relevant Courses:** Software Engineering, Data Structures (C++), Advanced Programming, Databases, Computer Systems, Blockchain, Discrete Mathematics, Probability & Statistics, Linear Algebra, Calculus